The Play by Nature

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This book should be read, can be read, by almost everyone. It describes with great skill a new face of the theory of evolution. With much of the light, unencumbered style that has lately sold new and sometimes erroneous biology to the public, it is, in my opinion, a more serious achievement. It succeeds in the seemingly impossible task of using simple, untechnical English to present some rather recondite and quasi-mathematical themes of recent evolutionary thought. Seen through this book in their broad perspective at last, these will surprise and refresh even many research biologists who might have supposed themselves already in the know. At least, so they surprised this reviewer. Yet, to repeat, the book remains easily readable by anyone with the least grounding in science.

Even without intention to be snobbish, reading a popular book in a field close to one's research interests almost forces one to tally errors: this example misapplied, that point left ambiguous, that idea wrong, abandoned years ago. This book had an almost clean sheet from me. This is not to say that there are no probable errors—that could hardly be the case in a work where speculation is, in a sense, the stock in trade—but its biology as a whole is firmly the right way up and its questionable statements are at least undogmatic. The author's modest assessment of his own ideas tends to disarm criticism, and here and there the reader finds himself flattered by a suggestion that he should work out a better model if he doesn't like the one given. That such an invitation can be made seriously in a popular book vividly reflects the newness of the subject matter. Strangely, there are indeed possibilities that simple ideas as yet untested may shortly resolve some old puzzles of evolution.

What, then, is this new face of evolution? To a certain extent it is like a new interpretation of Shakespeare: it was all in the script but somehow it passed unseen. I should add, however, that the new view in question was latent not so much in Darwin's script of evolution as in nature's and that our lapse of attention is more on the scale of 20 years than of a hundred. Dawkins starts, for example, from those variable helical molecules that we now know fairly well; Darwin knew not even about chromosomes or their strange dance in the sexual process. But even 20 years is quite long enough to cause surprise.

The first chapter broadly characterizes the phenomena the book seeks to explain and shows their philosophical and practical importance to human life. Some intriguing and alarming animal examples catch our attention. The second chapter goes back to the first replicators in their primeval soup. We see these multiply and elaborate. They begin to compete for substrates, to fight, even to lyse and eat one another; they hide themselves and their gains and weapons in defensive stockades; these come to be used for shelter not only from the tactics of rivals and predators but from the physical hardships of the environments that the replicators are increasingly enabled to invade. Thus they mobilize, settle, throw up bizarre forms, pour over the beaches, across land, and right on to deserts and eternal snows. Between such frontiers, beyond which, for long, life cannot go, the soup is poured and repoured millions of times over into an ever-stranger diversity of molds; at length it is poured into ant and elephant, mandrill and man. This second chapter concludes, concerning some ultimate descendant coalitions of these ancient replicators: 'Their preservation is the ultimate rationale of our existence … Now they go by the name of genes and we are their survival machines.'

Forceful and provocative, the reader may think, but is it very new? Well, so far perhaps not, but of course evolution has not ended with our bodies. More important still, the techniques of survival in a crowded world turn out to be unexpectedly subtle, much more subtle than biologists